

QFT

$$\mathcal{L} = \frac{1}{2} \left(\frac{d\phi}{dt} \right)^2 - \frac{1}{2} \left(\frac{d\phi}{dx} \right)^2 - m^2 \phi^2$$

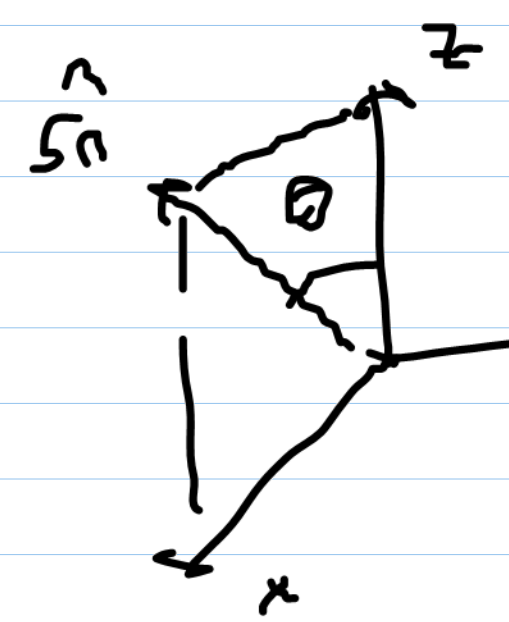
$$\Pi = \frac{\delta \mathcal{L}}{\delta (\partial_0 \phi)} \Rightarrow [\Pi(y, t), \phi(x, t)] = -i \delta(y-x)$$

$$\phi(x, t) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega p}} [\phi(p) e^{-i\omega p t + i p x} + \phi^*(p) e^{i\omega p t - i p x}]$$

$$[\phi^*(p_1), \phi(p_2)] = \delta(p_1 - p_2)$$

Rotational Sym

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$



$$\hat{s}_n = s_n \sin \theta + s_z \cos \theta$$

$$S_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$S_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\hat{s}_n = \sin \theta \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \cos \theta \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \Rightarrow$$

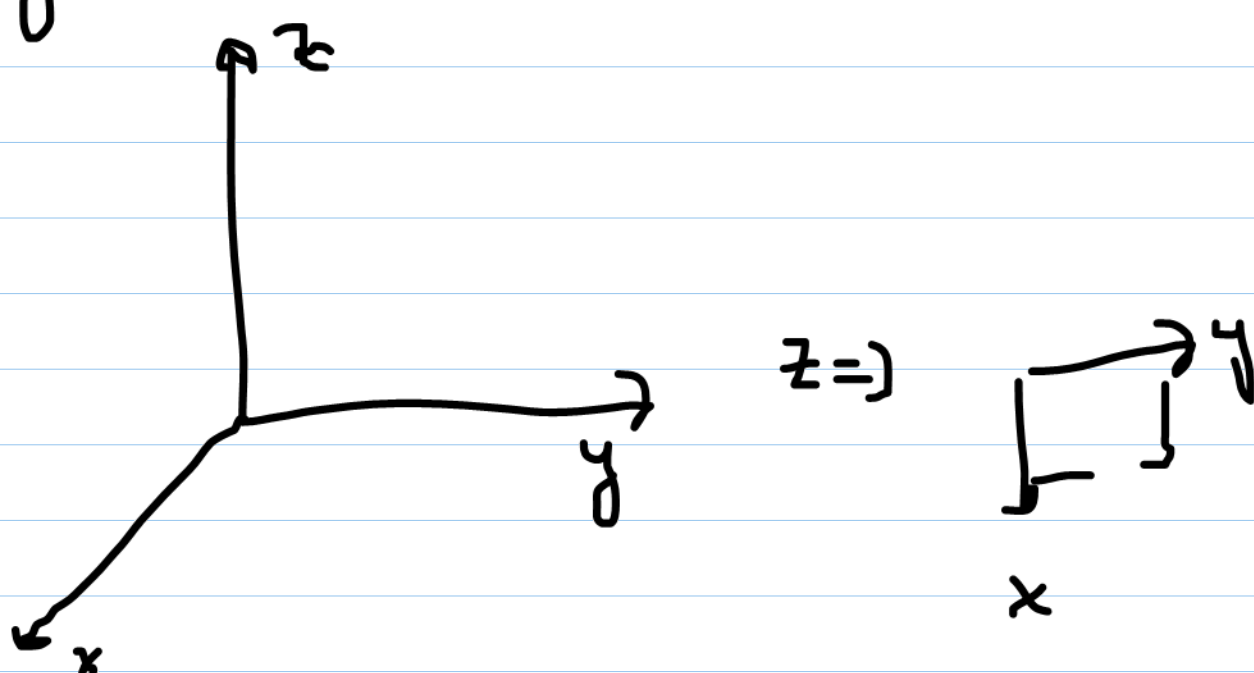
$$\hat{R}(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$$

$$R \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$|x, y|_{x^2+y^2=1} = x^2 + y^2 = 1$$

$$x^2 + y^2 = (x, y) \begin{pmatrix} x \\ y \end{pmatrix} = (x, y) \underbrace{\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}}_{\mathbb{1}} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$A \cdot A^{-1} = \mathbb{1}$$



$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = R_{xy}(\theta)$$

$$R_{yz}=? \quad R_{xz}=?$$

$$R_{yz}(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$$

$$R_{xz}(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$$

$$\underbrace{x^2 + y^2}_{\text{Rotational sym}} = x^2 + y^2$$

$$R \cdot \mathbb{1} = \mathbb{1}$$

$$R' \mathbb{1} = \mathbb{1}$$

Alan $\phi(x)$ rotational altında skaler old. için değişmez.

$$\phi'(x) = \phi(x)$$

Bu nedenle $\mathcal{L}$ da rotational altında değişmez.

vector karesi

$$V = \begin{pmatrix} V_x \\ V_y \\ V_z \end{pmatrix} \quad W = \begin{pmatrix} W_x \\ W_y \\ W_z \end{pmatrix}$$

skaler carpımı

$$\begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} V \\ W \end{pmatrix}$$

$$\text{Bu 2 vekt. arasındaki carpım } V^T W = V^T W = V_x W_x + V_y W_y + V_z W_z$$

Skaler çarpımın değişmezliği

$$(AB)^T = (A^T B^T)^T = A^T B^T A^T B^T = A^T A^T B^T B^T$$

$$(AB)^T = \sum_i (A^T B^T)_{ij} = \sum_j (A^T B^T)_{ji} = A^T A^T B^T B^T$$

∇ skaler funkt hareket yaratır.

$\nabla \phi = \begin{pmatrix} \frac{\partial \phi}{\partial x} \\ \frac{\partial \phi}{\partial y} \\ \frac{\partial \phi}{\partial z} \end{pmatrix}$ ∇ bir skaler funkt üzerinde (ϕ) etki edince bir vektör alanı yaratır. Buna gradyan denir.

$$L = \underbrace{\frac{1}{2} m \dot{\phi}^2}_{\text{kinetik}} - \underbrace{\frac{1}{2} (\nabla \phi)^2}_{\text{potansiyel}} - \underbrace{\frac{1}{2} m^2 \phi^2}_{\text{kütlesi terimi}}$$

skaler alan (ϕ) vektör alan complex skaler alan

skaler alan Her noktada tek bir sayı (örn sıcaklık yaygı)

Rot altında değişmez $\phi'(x) = \phi(x)$

vektör alan Her noktada yönü ve büyüklüğü olan (örn elektrik alan)

Complex skaler alan

$\phi(x)$ artık karmaşık $\phi(x) = \phi_1 + i \phi_2$ ϕ_1, ϕ_2 $U(1)$ hali scalar'dir.
(net, etkilenmez)

Boost Örn. Galilei Boost Lor. Boost

$$x' = (x - vt) \quad t' = t \quad y' = y \quad z' = z$$

$$x' = \gamma(x - vt) \quad t' = \gamma(t - vx) \quad y' = y \quad z' = z$$

$$\Lambda = \begin{pmatrix} \gamma & -\gamma v \\ -\gamma v & \gamma \end{pmatrix} \quad \text{sign } V \quad \Lambda = \begin{pmatrix} \gamma & -\gamma v \\ -\gamma v & \gamma \end{pmatrix}$$

$$t'^2 - x'^2 = t^2 - x^2 \quad \text{invariant olan büyüklük}$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \quad c=1 \Rightarrow \gamma = \frac{1}{\sqrt{1 - v^2}}$$

$$(t'^2 - x'^2) = \gamma^2 [(t - vx)^2 - (x - vt)^2] =$$

$$t^2 - 2txv + x^2 v^2 - x^2 + 2xtv - v^2 t^2 = x^2(v^2 - 1) + t^2(1 - v^2)$$

$$(1 - v^2)(t^2 - x^2) \gamma^2 = (t^2 - x^2)$$

Lorentz altında değişmez

Dört vekt

$$x^\mu = (t, x, y, z) \quad x_{,\nu} = g_{\mu\nu} x^\nu$$

indis indirme kaldırma

$$g_{\mu\nu} = \begin{pmatrix} 1 & -1 & -1 & -1 \end{pmatrix} \quad \vee \quad g^{\mu\nu} = \begin{pmatrix} -1 & 1 & 1 & 1 \end{pmatrix}$$

$$x'^\mu = \Lambda^\mu_\nu x^\nu \quad \Lambda^T g \Lambda = g \quad \text{Lorentz dön. tanımı}$$

Dort vekt. in compim

$$A \cdot B = ? \quad A \cdot B = A_\mu B_\nu g^{\mu\nu} = A^\mu B^\nu g_{\mu\nu}$$

$$A'^\mu = \Lambda^\mu_\nu A^\nu \quad B'^\mu = \Lambda^\mu_\rho B^\rho$$

$$A'^\mu g_{\mu\nu} B'^\nu = A^\alpha g_{\alpha\beta} B^\beta \quad \text{Lorentz invariance}$$

$$A^\mu = (A^0, \vec{A}) \quad A_\mu = (A_0, -\vec{A})$$

$$\Lambda = \begin{pmatrix} 1 & & & \\ & \gamma & & \\ & & \gamma & \\ & & & 1 \end{pmatrix}_{4 \times 4} \quad A'^\mu g_{\mu\nu} = A_\nu = \begin{pmatrix} 1 & & & \\ & \gamma & & \\ & & \gamma & \\ & & & 1 \end{pmatrix}_{4 \times 4} \cdot \begin{pmatrix} 1 & & & \\ & \gamma & & \\ & & \gamma & \\ & & & 1 \end{pmatrix}_{4 \times 4} = \begin{pmatrix} 1 & & & \\ & \gamma & & \\ & & \gamma & \\ & & & 1 \end{pmatrix}_{4 \times 4}$$

$$\Rightarrow (A^0, A_x, A_y, A_z) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \gamma & 0 & 0 \\ 0 & 0 & \gamma & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = (A^0, -A_x, -A_y, -A_z)_{4 \times 4}$$

$$A \cdot B = A^0 B^0 - \vec{A} \cdot \vec{B}$$

$$A_\mu B^\mu = g_{\mu\nu} A^\mu B^\nu = A^0 B^0 - A^1 B^1 - A^2 B^2 - A^3 B^3$$

$$\text{general } \mu, \nu \quad 0, \dots, 3$$

$$i, j \quad 1, \dots, 3$$

$$1 \rightarrow 2 \rightarrow 3$$

$$(\partial_\mu \phi)(\partial^\mu \phi) = (\partial_0 \phi)^2 - (\partial_1 \phi)(\partial_1 \phi) \quad L = T - V$$

$$L = \frac{1}{2} (\partial_\mu \phi_1)^2 + \frac{1}{2} (\partial_\mu \phi_2)^2 - \frac{1}{2} m^2 \phi_1^2 - \frac{1}{2} m^2 \phi_2^2 - V(\phi_1^2 + \phi_2^2)$$